

**RULES AND REGULATIONS FOR THE WRITTEN EXAMS**  
**AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES**

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your CELLPHONES AND SMART DEVICES; and PLACE THEM **away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is **VISIBLE FROM OUTSIDE** during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

**In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.**

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

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Student Full Name:

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Student Number:

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Course Code and Title:

**AIN-2002 INTRODUCTION TO DATA SCIENCE**

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Signature:

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Date:

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| Q.1 | Q.2 | Q.3 | Q.4 | Q.5 | TOTAL |
|-----|-----|-----|-----|-----|-------|
|-----|-----|-----|-----|-----|-------|

**Q.1) (15 pts)** Please answer the questions according to the exhibit:

```
set.seed(1)
scores <- c(rnorm(10, mean=60, sd=5), 99, 100, 101)
boxplot(scores, main="Student Scores")
z_scores <- scale(scores)
which(abs(z_scores) > 2.5)
```

**A) Please explain the role of scale() function in the context (5pts)**

**B) What is the role of abs(z\_scores) > 2.5? (5pts)**

**C) Based on the boxplot(scores) command, please draw the graph and clearly indicate the median, quartiles, and any outliers. Label the extreme values (5pts)**

## FINAL EXAM

(This is a 100-minute exam)

**Q.2) (20 pts)** Please answer the questions according to the exhibits:Input code:

```
sample <- c(67, 70, 72, 68, 71, 74, 69, 73, 75, 76)
t.test(sample, mu=70, conf.level=0.95)
```

Output Code:

```
t = 2.449, df = 9, p-value = 0.0378
alternative hypothesis: true mean is not equal to 70
95 percent confidence interval:  69.38  74.82
sample mean: 72.1
```

**a) (5pts)** Please define Null and Alternative Hypotheses**b) (5pts)** Interpret the **p-value** in this context. Where do we use it?**c) (5pts)** What does **confidence interval lower bound (69.38)** and **upper bound (74.82)** tell us?**d) (5pts)** Would your interpretation change if the **sample size** were 100 instead of 10? Explain why?

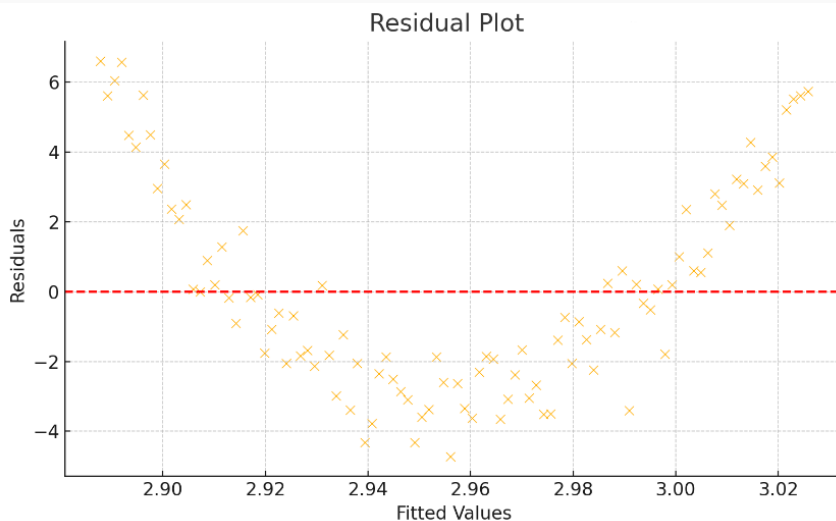
(This is a 100-minute exam)

**Q.3) (25pts)** We want to understand how a car's **engine power (hp)** and **weight (wt)** affect how many miles per gallon (mpg) it can drive. That is, how do heavier cars or more powerful engines change fuel efficiency. For that reason, we are trying to model the outcome as follows:

```
model <- lm(mpg ~ hp + wt, data = mtcars)
summary(model)
plot(model$fitted.values, model$residuals)
abline(h = 0, col = "red")
```

Output code:

```
Coefficients:
(Intercept)  37.227
hp           -0.032
wt           -3.878
```



Please answer the questions according to the exhibit:

**a) (5pts)** Please write the model as a mathematical equation.

## FINAL EXAM

(This is a 100-minute exam)

**b) (5pts)** Explain what it means when the coefficient **wt** is negative and statistically significant in the `summary(model)` output. What does this tell you about the relationship between car weight and fuel efficiency?

**c) (5pts)** What is the function of **abline** (**h = 0, col = "red"**) on residual plot? What does it tell us?

**d) (5pts)** Interpret the residual plot. What would that suggest about the model?

**e) (5pts)** Is the modelling method used in R correct? Yes/No and why?

**Q.4) (25pts)**

```
set.seed(123)
df <- data.frame(
  growth = c(rnorm(10, 5), rnorm(10, 6), rnorm(10, 8)),
  fertilizer = rep(c("A", "B", "C"), each=10)
)
fit <- aov(growth ~ fertilizer, data = df)
summary(fit)
TukeyHSD(fit)
```

**a) (5pts)** What does **aov()** function do.

**b) (5pts)** What does fit variable tell us in this context?

**c) (5pts)** When is it appropriate to use Tukey's HSD? What does it tell us?

**d) (5pts)** Given the p-value for A vs C Tukey's result was 0.0002. What conclusion would it tell you?

**e) (5pts)** Suppose the group means were as follows:

- a. Fertilizer A mean = 5.0
- b. Fertilizer B mean = 6.0
- c. Fertilizer C mean = 8.0

Use this information and your understanding of ANOVA to explain why the p-value for A vs C is so small.

**Q.5) (15pts)** Please answer the questions according to the exhibit:

```
library(tidyr)
library(dplyr)

survey <- data.frame(
  ID = 1:4,
  full_date = c("2023-01-15", "2023-03-22", NA, "2023-05-30"),
  satisfaction = c(5, NA, 4, 3)
)

cleaned_survey <- survey %>%
  separate(full_date, into = c("Year", "Month", "Day"), sep = "-", remove = FALSE) %>%
  replace_na(list(satisfaction = 0))
```

**a) (5pts)** What does the **separate()** function do in this code? What does it do to the **full\_date** column?

**b) (5pts)** What is the effect of **replace\_na(list(satisfaction = 0))**? In what types of data collection scenarios would this be a reasonable choice, and in which cases it could introduce additional problems?

**c) (5pts)** After running this code, one row has NA values for Year, Month, and Day. What does this indicate about the original **full\_date** column in that row? How should a data analyst respond to this kind of missing or incomplete data??